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Exam 3

- The exam is closed notes. No LLMs, search engines, notes, slides, videos, chats with friends, etc. are to be used on the test.
- All answers must be in your own words.
- You will receive a grade of zero if you copy or paraphrase from the class slides or textbooks, web search results, or LLMs.
- Please write legibly if you are writing answers by hand. Uploaded photos must be in a standard orientation & format (JPG, PNG, PDF) not HEIC/F.
- The exam is from 12:45 - 2 PM (except VISAs). Any questions answered after 2:05 will receive zeroes.



Submitted successfully!

Assignment submitted and ready for grading.

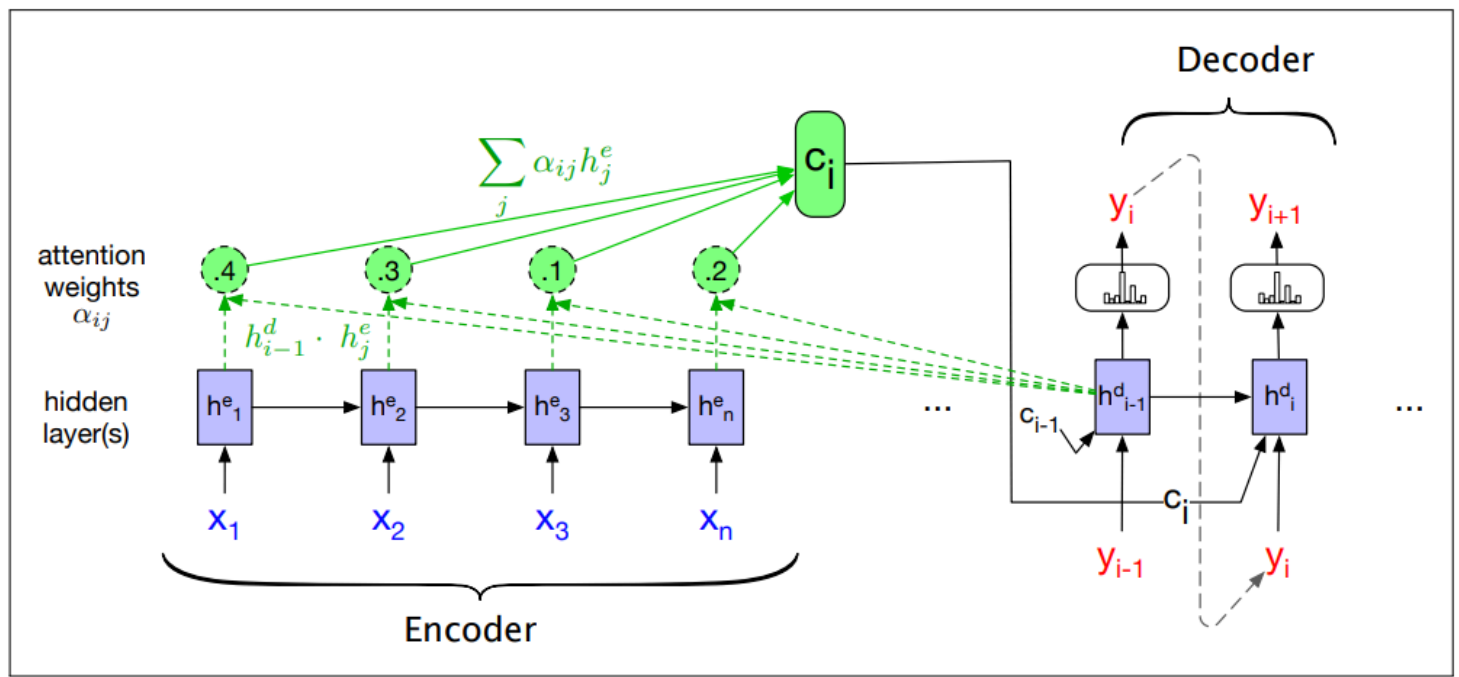
You can resubmit until **Thursday, April 9, 2026 3:00 pm (Eastern Daylight Time)** if needed.

[Edit and resubmit](#)

Q1A (9 points)

Submitted

The following figure shows an example Encoder-Decoder and the use of attention to compute a dynamic context vector.



In this figure, we are processing the input y_i to generate the output y_{i+1} . In your own words, (1) list the 3 input components needed to perform the hidden layer computation for this current decoding step (h_i^d) and (2) briefly explain each component.

(1) In this cross-attention mechanism, the 3 components are Q, K, and V.

(2) the query (Q) vector is comprised of the current and past tokens generated by the decoder, the key (K) vector is extracted from the encoders hidden states to be compared to Q, and the value (V) vector also from the encoder. The post-activation function with Q and K scales V in the final output.

Q1B (4 points)

Submitted

Using the Encoder-Decoder figure in Q1A, explain in terms of the *Language Modeling* task why the current decoding step h_i^d depends on calculations of the previous decoding step's hidden layer h_{i-1}^d and its relationship to the Encoder's hidden states $h_{1..n}^e$.
Hint: Remember in class I explained this in terms of the Machine Translation task.

Because the decoder's purpose is to predict the next token in a sequence. To reduce snooping behavior during training, the decoder only relies on the relationships learned from the encoder given the decoders current tokens.

Q2 (3 points)

Submitted

List the three different pieces of neural architecture found in LSTM, Transformer, and Encoder-Decoder that are mathematically similar (ignoring attention) and functionally equivalent.

(1) Activation functions, (2) linear-combinations resulting in z-values, (3) ...

Q3A (4 points)

Submitted

For RNN trained for language modeling, we saw that we can ignore the output of the hidden layer at each time step, and use only the output of the final hidden state as input to a classifier that is fine-tuned for a different downstream prediction task.

For the Encoder-Decoder: (1) What replaces that final hidden state output acting as input to the next classifier? (2) What piece of the model acts as the next classifier?

(1) A context vector as an output of the encoder. (2) The decoder acts as the next classifier. Especially for word prediction.

Q3B (3 points)

Submitted

For RNN trained for language modeling, we saw that we can ignore the output of the hidden layer at each time step, and use only the output of the final hidden state as input to a classifier that is fine-tuned for a different downstream prediction task.

For the bidirectional transformer used as a pretrained language model (PLM), specifically BERT: (1) What replaces the final hidden state output that will be used as input to the next classifier? (2) Any classifier could be used for downstream prediction, but what do BERT-based PLMs typically use instead?

(1) Post-self-attention output (2) Feed forward neural network

Q4A (2 points)

Submitted

When constructing a Decision Tree, the information gain from a split is calculated as 0.3, but when you implement 5-fold cross-validation, you notice that the model performs poorly on test data despite high training accuracy.

What phenomenon is likely happening here?

- ☐ Underfitting
- ☒ Overfitting
- ☐ Imbalanced Dataset
- ☐ Balanced Dataset

Q4B (5 points)

Submitted

When constructing a Decision Tree, the information gain from a split is calculated as 0.3, but when you implement 5-fold cross-validation, you notice that the model performs poorly on test data despite high training accuracy.

Explain one characteristic of the Decision Tree algorithm that could cause this phenomenon and how you would fix it.

By implementing more splits in the Decision Tree, more branches are created, fitting to noise instead of meaningful patterns. This prevented my "pruning" branches, i.e., removing inconsequently splits.

Q5 (8 points)

Submitted

What is the main simplifying assumption of the Naive Bayes algorithm and why is it *mathematically* necessary?

That each of the probabilities of a certain outcome is "conditionally independent" of all other outcomes. While in real-life this is almost always false, it is useful because we can reliably rank the probability of observing a particular class relative to other classes. Specifically, the formula goes $P(x_1, x_2 | C) = P(C) * P(C | x_1) * P(C | x_2)$.

Q6A (6 points)

Submitted

List 2 limitations of the k -means algorithm. For each, briefly explain how the algorithm contributes to these limitations.

(1) With improper initialization, centroids get pulled to local optima instead of global optima, obscuring the separation. In other words, the algorithm contributes to poor clustering when the centroid locations are improperly selected.

(2) The number of centroids must be defined ahead of time. When there are too few centroids, the algorithm attempts to capture data points that may belong in another cluster in reality.

Q6B (2 points)

Submitted

Knowing that k -means struggles when clusters have irregular shapes (i.e., more elliptical or other complex shapes instead of circular), what assumption is the algorithm making about the data?

That the data is spherical by nature

Q7 (0 points)

Not answered

Please carefully read the following Problem Description. You will answer for each model in their respective question boxes. Please justify your choices clearly and succinctly.

Problem Description: You are given a dataset of 1 million tweets, each with 10 useful features (e.g., day of tweet, time, geolocation, number of retweets, etc.) as well as the original tweet text. You have the true labels for each tweet. Assume the data is from a Gaussian distribution and is mostly linearly separable with less than 5% of samples being outliers. The prediction task can be binary or probabilistic as needed to work with the model. Your goal is to predict if the tweet will go viral or not.

For each model, briefly explain how well that model could work on this dataset for this prediction (if it can). Note if the model could only be used if certain conditions hold or there is anything you would need to change. Be sure to write down any assumptions you make. You should include anything that helped you make your decision, e.g., if you would use the entire dataset and feature set, if you would make any modifications, if you would do any pre-processing, kernel choice, architectural choices, etc.

The models are: (a) Decision Tree, (b) k -means clustering, (c) Naive Bayes, (d) Support Vector Machine (SVM), and (e) Transformer.

You can answer for each model separately in the following boxes.

Q7A (5 points)

Submitted

Decision Tree

Since Decision Trees are a supervised learning technique used for both regression and classification, it would work well for predicting binary outcomes (tweet goes viral or not). Especially when the data is mostly linearly separable.

Q7B (5 points)

Submitted

k -means Clustering

K-means is an unsupervised learning technique, so right away there's some misalignment between the model and the tweet classification task. However, if you selected only numerical variables (e.g., time, geolocation, number of retweets) used for euclidean distance and limited to two centroids (one for viral and one for not viral), you could predict the status of future tweets based on their position in either training cluster.

Q7C (5 points)

Submitted

Naive Bayes

Naive Bayes is well suited for categorical predictions such as this. You would need to select for cardinal features in the data or discretize the numerical features. This allows NB to rank the probability of a tweet going viral or not given a row of conditions.

Q7D (5 points)

Submitted

Support Vector Machine (SVM); with your choice of kernel from: linear, polynomial, sigmoid, RBF (Gaussian)

An SVM would definitely work for binary classification given that the inputs are encoded properly with an RBF kernel (could be linear since the data is linearly separable). For example, target-encoding numeric features and one-hot encoding binary cardinal features.

Q7E (5 points)

Submitted

Transformer; with your choice of simple or bidirectional Transformer, and your choice of type of attention

Transformers are seq2seq models conventionally. Given that viral/not viral is a binary prediction, other models are better suited.

Q8 (5 points)

Submitted

What is the relationship between generalization, the shape of the data (sample size & dimensionality), and overfitting?

With low sample size, high training performance, and low test performance, the model is likely overfit. The model is training to fit a small sample of data relative to the entire population (i.e., it is extrapolating poorly). Having more features allows the model to look at the same sample size with more perspectives (i.e., dimensions), improving generalizability.

Q9 (2 points)

Submitted

Choose the answer which best fills in the blanks to make a true statement: _____ is one technique which can improve _____ by controlling _____.

- ☐ Regularization, overfitting, generalization
- ☒ Regularization, generalization, overfitting
- ☐ Kernel implementation, performance, regularization
- ☐ Kernel implementation, performance, generalization

Q10 (2 points)

Submitted

Which of the following represents a source of bias that can affect our machine learning algorithms and models?

- ☐ Personal opinions and prejudices of annotators (people who label the data)
- ☐ Information sources that the data is collected from and how this data is collected
- ☐ Choosing the wrong algorithm/model for the dataset/task
- ☐ Removing data that causes results that do not match preconceived expectations
- ☒ All of the above

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